

EXPERIMENT 29. Bar Chart with Different Colors

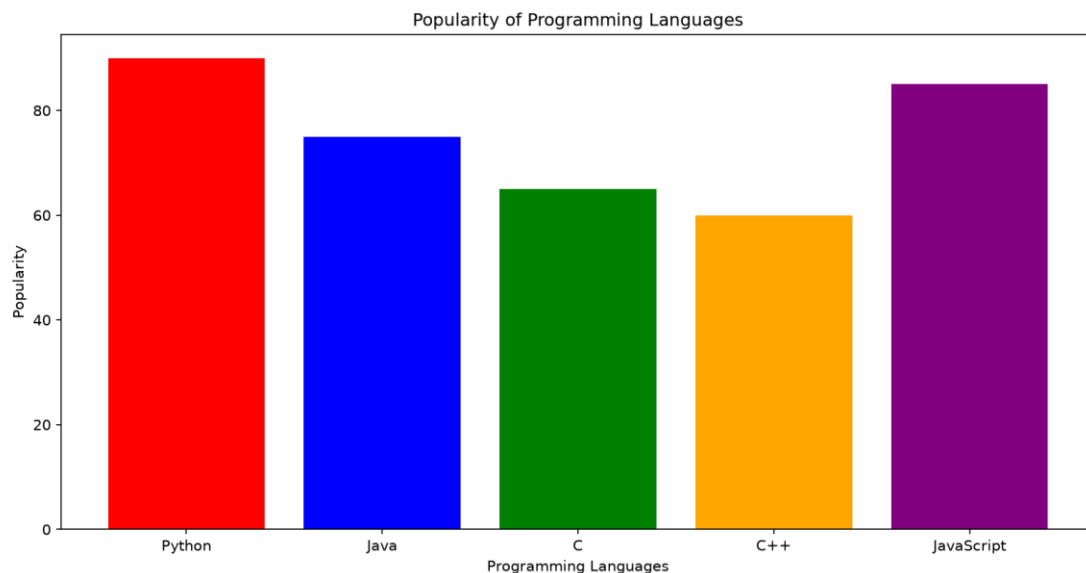
AIM:

To write a Python program to display a bar chart of programming language popularity using different colors.

CODE:

```
exp29.py > ...
1 import matplotlib.pyplot as plt
2 languages = ['Python', 'Java', 'C', 'C++', 'JavaScript']
3 popularity = [90, 75, 65, 60, 85]
4 plt.bar(
5     languages,
6     popularity,
7     color=['red', 'blue', 'green', 'orange', 'purple']
8 )
9 plt.xlabel("Programming Languages")
10 plt.ylabel("Popularity")
11 plt.title("Popularity of Programming Languages")
12 plt.show()
13
```

OUTPUT:



RESULT:

Thus, a multicolored bar chart was successfully created.

EXPERIMENT 30. Bar Plot of Scores by Group and Gender

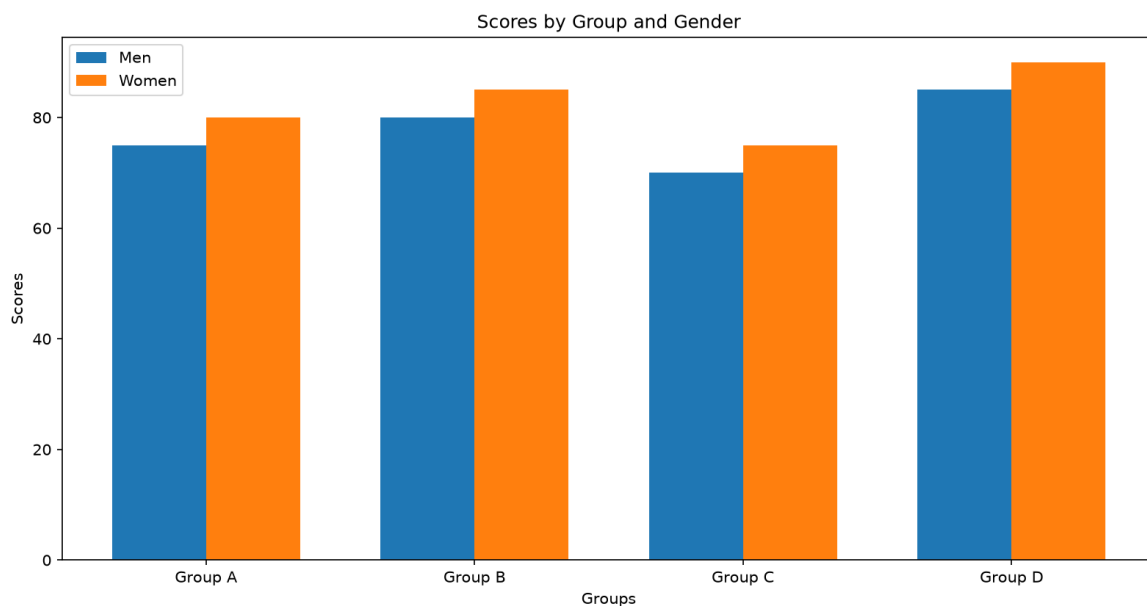
AIM:

To write a Python program to create a bar plot of scores by group and gender using multiple X-values.

CODE:

```
exp30.py > ...
1  import matplotlib.pyplot as plt
2  import numpy as np
3  groups = ['Group A', 'Group B', 'Group C', 'Group D']
4  men = [75, 80, 70, 85]
5  women = [80, 85, 75, 90]
6  x = np.arange(len(groups))
7  width = 0.35
8  plt.bar(x - width/2, men, width, label='Men')
9  plt.bar(x + width/2, women, width, label='Women')
10 plt.xlabel("Groups")
11 plt.ylabel("Scores")
12 plt.title("Scores by Group and Gender")
13 plt.xticks(x, groups)
14 plt.legend()
15 plt.show()
```

OUTPUT:



RESULT:

Thus, the scores by group and gender were successfully represented using a grouped bar chart.